CELIA RUBIO-MADRIGAL

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PhD researcher in machine learning at CISPA, with experience in algorithm development, mathematical analysis, and empirical evaluation. My research investigates how graph structure and feature design influence learning performance, with an emphasis on capturing long-range dependencies. Expected PhD completion: late 2027.

Experience

- **Doctoral researcher** – *CISPA Helmholtz Center for Information Security* **2023-27 (expected)**
  - Empirical evaluation: Systematic hyperparameter sweeps and ablation studies of graph-agnostic models trained on graph-derived features in PyTorch, matching or outperforming state-of-the-art GNNs (ICML 2026).
  - Algorithm development and mathematical analysis: Developed graph-rewiring approaches and theoretical analyses of mean aggregation, graph spectra, and community structure (ICLR 2025; NeurIPS 2024).
  - Advisor: Dr. Rebekka Burkholz at the Relational Machine Learning Group
- **RA: Departmental collaboration** – *Ministerio de Educación & Universidad Complutense de Madrid* **2021-22**
  - Identifying complex Boolean functions with ML – Advisor: Dr. Ismael Rodríguez Laguna
- **REU: Introduction to research ICMAT-Severo Ochoa** – *Instituto de Ciencias Matemáticas (ICMAT)* **2021**
  - Differential geometry methods in ML applications – Advisor: Dr. David Martín de Diego
- **Intern: Consultant Developer** – *Guidewire Software* – Built an engine for retrieving internal issues. **2021**

Education

- **PhD in Computer Science** – *CISPA & Universität des Saarlandes (Saarbrücken, Germany)* **2023-27 (expected)**
- **MSc in Advanced CS with AI** – *University of Strathclyde (Glasgow, UK)* – 85.7/100 (Distinction) **2022-23**
  - Recipient of the "la Caixa" Postgraduate Fellowship – Department's Best Overall Performance Prize
- **BSc in Mathematics** – *Universidad Complutense de Madrid (Spain)* – 9.08/10 **2017-22**
- **BSc in Computer Science** – *Universidad Complutense de Madrid (Spain)* – 9.16/10 **2017-22**

Publications

Legend: * Equal contribution. † Co-supervision.

Joshi, Gadhikar†, Rubio-Madrigal†, Dutta, Singh, Burkholz. *Prune to Protect: Faster Training and Enhanced Privacy by Dynamic Data Pruning.* NeurIPS'26.

Rubio-Madrigal, Burkholz. *Fixed Aggregation Features Can Rival GNNs.* ICML'26.

Reddy, Verma, Rubio-Madrigal, Muandet, Burkholz. *Boosting for Predictive Sufficiency.* ICLR'26.

Reddy, Rubio-Madrigal, Burkholz, Muandet. *When Shift Happens - Confounding Is to Blame.* ICLR'26.

Jacobs, Gadhikar, Rubio-Madrigal, Burkholz. *Hyperbolic Aware Minimization: Implicit Bias for Sparsity.* ICLR'26.

Rubio-Madrigal*, Jamadandi*, Burkholz. *GNNs Getting ComFy: Community and Feature Similarity Guided Rewiring.* ICLR'25.

Jamadandi*, Rubio-Madrigal*, Burkholz. *Spectral Graph Pruning against Over-squashing and Over-smoothing.* NeurIPS'24.

Awards

– 13th Heidelberg Laureate Forum: *200 international young researchers network with math and CS laureates* **2026**
– Dept. Best Overall Performance Prize: *University of Strathclyde's top student in MSc Advanced CS* **2023**
– 🔗 Nova 111 Student List: *Spain's top 10 talents under 25 in Computer Science* **2023**
– 9th Heidelberg Laureate Forum. Featured in: *HLFF Spotlight*, *Tribuna Complutense* **2022**
– 🔗 "la Caixa" Fellowship: *Full funding for MSc studies.* Featured in: *El País*, *El Plural*, *TeleMadrid* **2022**

- **Undergraduate**

– 🔗 Aula de Verano "Blas Cabrera" for Spain's top undergrad records: *Intl. Univ. Menéndez Pelayo (UIMP)* **2022**
– 🔗 SWERC contestant (competitive programming): *ICPC Southwestern Europe Regional Contest in Italy* **2022**
– Becas de Excelencia: *Excellence Grants for undergrads with >9/10 in Madrid (2100€)* **2018-22**
– 🔗 EGMO contestant representing Spain: *European Girls' Mathematical Olympiad in Switzerland* **2017**

Talks

- **Invited talks** – *Seminars and reading groups* **2025-26**

– GLOW 🔗, TUWien ML RU seminar 🔗, Cohere For AI ▶, CISPA ML seminar, AI Security Afternoon 🔗.

- **Contributed talks** – *Conferences and workshops* **2024-26**

– GSP 🔗 2026, HAICON 🔗 2024. Lightning talks: GTML 🔗 2025, MLG 🔗 2025 (2), NetSci 🔗 2025.

- **Non-archival posters** – *Non-archival conferences, workshops, and sessions* **2024-26**

– GFM 🔗 2026, GSP 🔗 2026, CPAL 🔗 2026 (3), WiML 🔗 2024 🔗 2025.

Teaching

- **Math Olympiad trainer** – 🔗 *International EGMO Preparatory Seminar in Bonn, Germany* **2026**
- **Master's and Bachelor's thesis advising** – *CISPA & Universität des Saarlandes* **2024-present**
- **Teaching Assistant: Laboratory demonstrator (7h/week)** – *University of Strathclyde* **2023**

– "CS310 Foundations of AI" for Year 3 BSc Computer Science – Lecturer: Dr. Andrew Abel

Service

- **Academic service** **2024-present**

– Reviewer: MLG 2025-26, ICLR 2026 – **Top 200 (1%)**, CPAL 2026, ICML 2026 – **Gold (25%)**, NeurIPS 2026.
– Volunteer at 19th and 20th Women in Machine Learning Workshop (WiML) at NeurIPS 2024 and 2025.

- **Math Olympiad service**

– 🔗 IT support at European Girls' Mathematical Olympiad (EGMO) with Joseph S. Myers **2023-present**
I manage technical operations for international juries and contestants: administer live contest Q&A, process multilingual problem translations, maintain internal communications, and log marks during coordinations.
– Deputy Leader of Spain at EGMO 2022 and PAGMO 2021. **2021-22**
– Jury at the local phases of 57th to 61st Olimpiada Matemática Española (OME) Madrid. **2020-24**

- **Student representation** – *Facultad de Informática (FdI), Universidad Complutense de Madrid* **2020-22**

– Secretary of the FdI Student Council 2021-22. Member of the FdI Faculty Board 2020-22.

- **LaTeX package development** – *Included in MiKTeX and TeXLive*

– 🔗 `mousik`: Music rendering for small excerpts with TikZ. **2026**
– 🔗 `ddphonism`: Twelve-tone music notation diagrams. **2019**